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Research Paper

Data-driven diagnostics of boiling heat transfer on flat heaters from non-intrusive visualization

Vinicius K. Scariot [a,1], Gustavo M. Hobold [b], Alexandre K. da Silva [a,*]

[a] Department of Mechanical Engineering, Federal University of Santa Catarina, Florianopolis, SC 88040-900, Brazil
[b] Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA 02139

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ABSTRACT

Rankine-based power generation cycles require the use of boiling heat exchangers for liquid–vapor phase change. However, operating boilers at too high heat fluxes can lead to heater meltdown due to onset of the boiling crisis. Despite recent advances in increasing this critical heat flux, the lack of fast and accurate diagnostics hinders boiling heat transfer safety at high heat fluxes. Here we show that neural networks fed on direct observation of bubble dynamics can be used to measure heat flux during nucleate boiling. The boiling heat transfer data was obtained with a horizontally oriented, circular flat heater with saturated water as the heat transfer fluid. A deep convolutional neural network (CNN) was trained using frames from videos acquired from 15 different heat dissipation values on the boiling curve to learn how these images correlate to the underlying heat flux. The CNN predicted heat fluxes with a mean square error of 14.6 W/cm^2. Furthermore, the CNN presented high sensitivity to pixel intensity in image regions relevant to boiling heat transfer physics (e.g. bubble boundaries, contextual clues about bubble image distortion). Understanding this approach may allow future studies in unstable and critical boiling regimes that require fast transient heat flux diagnostics and cannot rely on slower thermohydraulic measurements.

1. Introduction

Due to its ability to transport heat at high power densities (>1 MW/m^2) at minimal temperature differences (~15 °C in water), liquid–vapor phase change heat transfer finds application in nearly all industrial, grid-scale energy power plants. As such, Rankine power cycles, which rely heavily on boilers and condensers, are responsible for nearly half of the electrical power produced in the USA [1]. However, despite its relevance, heat fluxes associated with phase-change heat transfer, and in particular boiling heat transfer, are challenging to predict, model and measure accurately [2,3]. Historically, this has led to the reliance on the use of closed-form analytical expressions fitted from experimental results that predict heat flux from heater temperature, but which can deviate by up to ~ 100 % from experimental data [4]. Due to the intrinsically complex interplay between fluid mechanics, heat and mass transport, numerical simulations are often limited to few-bubble calculations in simple geometries or simpler modeling [5,6]. In addition to these challenges, boiling heat transfer is prone to catastrophic failure due to the onset of a boiling crisis, which, if not diagnosed, can cause an abrupt temperature increase on the heater, from ~ 150 °C to > 1000 °C on the timescale of seconds, depending on the system, engendered from the sudden formation of a vapor film that insulates the heater from the bulk liquid, potentially leading to catastrophic heater meltdown [7]. Accurate measurements in boiling experiments are often based on the use of high-speed cameras, infrared thermometry and particle image velocimetry (PIV) [8–10], which are not only expensive experimental techniques, but also involve substantial computational post-processing. On the other hand, simpler approaches such as extrapolation of Fourier conduction from temperature measurements are only accurate during pseudo-steady-state periods. This technique is limited by thermal inertia of the heater, which can impart significant delays to accurate heat flux and temperature measurements. Improving boiling diagnostics improves process safety of phase-change heat exchangers and can potentially increase the power density of heat transfer devices, especially at extreme heat flux conditions that are limited due to safety concerns.

Machine learning-based approaches to heat transfer prediction and diagnostics has recently received significant attention by the energy community, particularly as an alternative to heat transfer correlations to

* Corresponding author.
E-mail address: a.kupka@ufsc.br (A.K. da Silva).
[1] Department of Petroleum Engineering, University of Tulsa, Tulsa, OK 74104.

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predict heat flux as a function of thermohydraulic data (e.g., temperature, pressure, mass flow rate, among others) [11–13]. Our group demonstrated for the first time that computer vision and machine learning can be used to directly analyze boiling heat transfer from bubble visualization, demonstrating that visualization-based algorithms can perform sub-second detection of the onset of film boiling [14,15], accurately measure boiling heat flux purely from visualization over thin, μm-thick wires [16], without any reliance on thermohydraulic data. Soon after, Ravichandran *et al.* demonstrated that properly trained data-driven machine learning models can extract bubble diagnostics reliably from infrared thermometry, enabling quasi-real-time quantification of bubble period, growth time and nucleation site density [8,9]. The same group used transfer learning on a pre-trained CNN to study several dynamic parameters associated with sub cooled boiling [17], including the occurrence of dryout areas within the boiling surface [18]. Refs. [19] and [20] focused on studying the critical heat flux; the former considered the impact of analyzing images from different experimental setups for training and testing. These have motivated emerging follow-up studies exploring the use of machine learning to unravel boiling heat transfer diagnostics, demonstrating that heat flux can be predicted from thermoacoustics data of bubble dynamics [21,22], and that visualization-based tools can leverage transfer learning in data scarce scenarios to boiling chamber setups other than those on which it was originally trained [23]. Recently, quantification of boiling heat flux has also been shown possible in a physics-reinforced framework, in which a machine learning algorithm is used to extract bubble statistics, which is then used as a parameter to predict heat flux [24,25]. While such a framework leads to excellent quantification at low-to-moderate heat fluxes and allows the establishment of a causal relationship between heat flux and bubble dynamics, increasing the algorithm's reliability, it is also constraining in that it is only allowed to correlate specific bubble metrics (e.g., bubble size and count) with heat flux. Such approaches may fail near the critical heat flux (>100–130 W/cm^2 for water), which was recently shown to result from a complex bubble percolation at the heater [26,27], the dynamics of which is not necessarily captured by conventional bubble statistics.

Here we demonstrate that convolutional neural networks (CNNs) can be trained to infer heat flux directly from visualization, without explicit use of bubble statistics, supporting thermohydraulics data or expensive high-speed cameras. We develop our approach around a flat circular heater. The images are obtained at low speed (30 Hz) and processed at low resolution, reducing computational burden while retaining quantitative accuracy. Despite their simplicity, flat heaters can create complex, three-dimensional flow geometries with significant out-of-focus bubble-to-bubble interaction dynamics. Frames are acquired with a single DSLR camera from a conventional boiling setup with a copper block heated by cartridge heaters. The CNNs are trained solely on heat flux and visualization data. The study focuses on two novelties associated with the use of images to infer quantitative information from complex phase change processes. First, we show that the trained CNNs appear to extract information from the images that are relevant to boiling heat transfer physics. The CNN presented a high sensitivity to pixel intensity in image regions at bubble boundaries and areas presenting contextual clues about bubble volume distortion in the image. Second, it is shown that CNNs are able to interpolate between training conditions and estimate data in intermediate heat fluxes at which it was not trained; a mean square error (MSE) 14.6 W/cm^2 is reported. Finally, we suggest that CNNs may be able to accurately determine heat flux even in fast unsteady regimes that are otherwise inaccessible by conventional instruments like thermocouples if properly trained. One example of such is the transition boiling past the critical heat flux, where catastrophic heater meltdown may occur.

2. Experimental methods

The experimental setup used in this paper consists of a conventional pool boiling chamber (Fig. 1**a**) - further details of the boiling setup can be seen in **Figure S1**. This setup has been partly used in previous communications by our group, e.g. [15,16,28], using a wire as a heater [4] but, for the present study, its bottom surface was redesigned, fabricated and instrumented to be fitted with a flat heater. The flat heater or heater assembly was machined from a single electrolytic copper block, resulting in a disc with 30 mm in diameter, which was in contact with the heat transfer fluid. To avoid the direct contact between the heater assembly and the redesigned stainless steel bottom plate, a PTFE disk was used as shown in **Figure S1**. The heater assembly was fitted with eight 180 W cartridge heaters (their dimensions are shown in **Figure S1**), which were electrically connected in parallel configuration and powered by a 1500 W programmable DC power supply. Voltage drop over the cartridge heaters V_{heater} was measured with the help of a voltage divider circuit and the total current i_{total} was measured by the voltage drop V_{shunt} over a $R_{shunt} = 1$ mΩ current sensing shunt resistor ($i_{total} = V_{shunt}/R_{shunt}$). The total heat flow to the heater assembly was then inferred by the dissipated power of each individual heater (V^2_{heater}/i_{total}). The uncertainty over the voltage measurement was negligible (<mV), such that the uncertainty on the total heat flow was dominated by the uncertainty over the resistance of the shunt (0.5 %), leading to an uncertainty of ~ 0.5 % on the total heat flow measurement. Each cartridge heater was also instrumented with a Type J thermocouple. These were only used for monitoring the cartridge heaters temperature values and prevent overheating. Since these measurements were not used in the proposed analysis, the respective Type J thermocouples were not calibrated and presented a standard deviation of approximately ± 1 °C, which is satisfactory for their purpose.

The total power dissipated by the heaters is partially transferred to the surface and partially lost to the environment. Due to the inherent difficulty in determining all the heat losses, the present study relied on the Fourier law to calculate the heat flux dissipated to the working fluid (*i.e.*, deionized water) by the boiling surface. In this case, one needs temperature measurements in distinct vertical locations of the heater assembly. Three 2-mm diameter holes were drilled from the bottom surface of the heater assembly parallel to the cylinder axis. The center of these holes was located 4 mm from the axis, each located 120° apart. A Type T thermocouple was inserted in each hole and six-point calibrated between 50 °C and 140 °C to yield measurement errors lower than ± 0.1 °C. The thermocouples were 2 mm, 7 mm, 12 mm away from the boiling surface. Thermal paste was used to enhance the thermocouples' thermal contact with the heater assembly. From these temperature measurements, three estimates for the boiling surface temperature could be inferred. Applying a classical error propagation analysis [29], the heat flux determined through the Fourier law will mostly be affected by the uncertainties of the temperature measurements and the locations of the thermocouples, leading to an approximate overall uncertainty of ± 5 %; according to the supplier, the thermal conductivity of the copper material used is 389.1 W/mK. The uncertainty of the excess boiling surface temperature has a propagated uncertainty of ± 25 %; one should notice that the surface temperature was not used for training the CNN.

It is important to mention that the geometric features and dimensions of the heater assembly discussed above in **Figure S1** were obtained through a parametric numerical study using steady-state simulations in Comsol Multiphysics® [30]. The model only considered the heater assembly (*i.e.*, the copper block), which was subject to a series of boundary conditions aiming to mimic the actual heater. The goal was to guarantee that the geometry of the heater, as well as measurement locations, were sufficient such that heat flux and surface temperatures could be extrapolated from Fourier's law without significant loss of accuracy, and that the heat flux distribution at the surface was uniform. Therefore, the main variables considered in the parametric study were the diameter of the boiling surface, the height of the bottom and top cylinders composing the heater assembly and the location of the thermocouples within the heater assembly. For instance, the analysis tested four values for the boiling surface diameter: 50 mm, 40 mm, 35 mm and

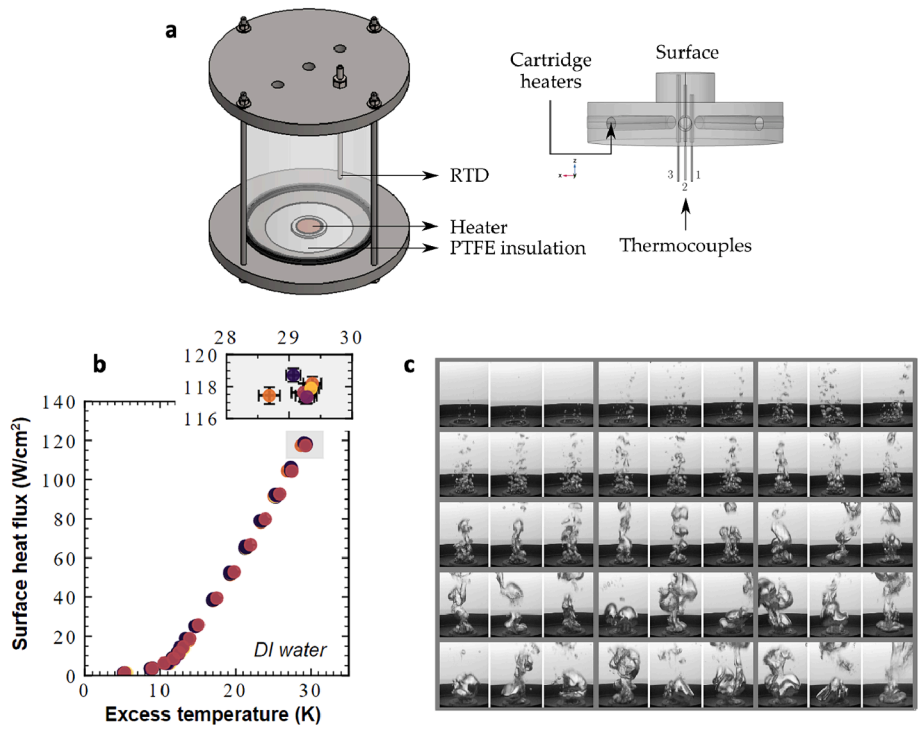

Fig. 1. Experimental details and reproducibility of the boiling setup. (a) Schematic representation of the experimental setup used for boiling heat transfer measurements and visualization. (b) Boiling heat transfer curves displaying heat flux *vs.* surface temperature for all experiments, and for heat fluxes computed from different combinations of thermocouples. Error bars represent the standard deviation of the temperature and heat flux within each heating step. (c) Representative downscaled, greyscale and cropped frames used as inputs to the machine learning algorithm along with their respective heat fluxes.

30 mm. For a diameter of 50 mm, the boiling surface would not reach the critical heat flux, even if the heater were subjected to the highest available power dissipation. As for 40 mm and 35 mm, the actual critical heat flux could actually be reached, however, there would be a small safety margin as the heaters would operate too close to the maximal power dissipation. Therefore, a 30 mm boiling surface diameter was selected. As for the height of the boiling surface (see **Figure S1**), three values were tested and it was verified that 15 mm was the best compromise solution. This height affects the heat flux experimental uncertainty, its uniformity and the temperature difference throughout the heat assembly. The height of the heater assembly, which housed the heat cartridges, presented a less significant effect on the heater flux uniformity and on the overall temperature difference across the heater assembly. In this case, the height of the base was set as 20 mm. Finally, a parametric study was also implemented for the location of the thermocouples, while considering different aspects, including machining limitations and the effect of the thermal paste.

The experimental setup was also instrumented with a resistance temperature detector and a pressure transducer, allowing measurements of the fluid bulk temperature and of the atmospheric pressure. The copper surface was progressively polished using sandpaper, with grit values increasing from #80 to #400, resulting in mean surface roughness, which was measured in five different locations of the heater and varied from 0.353 to 0.665 μm (ISO 25178, ZYGO Laser Interferometer). The maximal static and minimal receding angles of contact on the surface were measured to be 97.4° and 13.7°, respectively using the Drop Shape Analyzer optical goniometer from Krüss as shown in **Figure S2**. The heater assembly is shown in Fig. 1.

The boiling chamber was filled with deionized water before each boiling curve was recorded, and an 800 W auxiliary heater was used to bring the bulk temperature to saturation before beginning each experiment. Then, the cartridge heaters were set to dissipate heat rates varying from 20 W to 900 W (1.2 W/cm^2 to 118 W/cm^2), in 15 steady state steps. For each heat flux, at least 60 s of data was recorded (*i.e.*, temperatures, pressure, power, and video). In total, seven boiling curves were captured (Fig. 1**b**), all of which showed excellent agreement with each other with respect to heat flux and temperature measurements. The first three tests were used for training, the following two for validation and the remaining two for testing. Finally, all videos used in the training and comparison of the results provided with the machine learning algorithm were acquired using an over-the-counter Nikon D3100 with 30 mm to 50 mm lens at 30 Hz frame rate and a 720p resolution (1280 × 720 pixels per frame). The use of a conventional 30 Hz camera rather a high-speed camera has been shown to be appropriate [14]. Typically, high-speed cameras are used in boiling heat transfer studies to produce highly correlated images allowing one to track parameters of interest and study the system's dynamics. However, the current study does not analyze the system dynamics, and the trained CNN predicts the heat flux based on a single image. Hence, a lower frame rate of 30 Hz was selected to avoid overfitting. Lighting was provided by back-mounted LEDs through a light diffuser.

3. Machine learning models

Machine learning models are more prone to fail when dealing with highly-dimensional data [31], and given that colored high resolution frames contain three times more degrees of freedom, the dimensionality of the data was further reduced by transforming each frame to greyscale, downscaling the resolution using a nearest-neighbor algorithm and cropping out parts of the image that were not related to boiling heat transfer. In addition, to reduce position bias on each image, the crop window was shifted by ± 3 % for each frame during training. This is an important step since machine-learning algorithms might focus on specific image details that are unrelated to the phase change physics and use that to make predictions. For instance, the relative position of the camera and lighting with respect to the boiling surface is constant for each experimental condition but not necessarily identical from one condition to another. This unintentionally added spurious correlation in

the data could potentially be used to predict the heat flux from the image. The random shifting reduces this effect.

The resulting frames contain 96 × 128 pixels, or 12,288 degrees of freedom. Examples of the resulting images for each heat flux are shown in Fig. 1c for two boiling heat transfer regimes: natural convection and nucleate boiling. This figure frame, which is shown in a greyscale, present three distinct images for each of the fifteen heat dissipation levels. The heat dissipation increases from left to right and from the top to the bottom of Fig. 1c. For instance, the three first columns of the upper row highlights images within the natural convection regime, while the three last columns of the last row indicated images within the nucleate boiling regime.

The heat flux values were also preprocessed before being used in the machine learning algorithm. This was done by calculating the mean value and standard deviation for the training dataset and subsequently subtracting each label value by the training mean and dividing by the training standard deviation. Hence, the target values were normalized.

Machine learning models in supervised learning are parameterized mathematical entities that transform a numerical input into a numerical output, both of which can have arbitrary dimensions depending on the application. These models may contain a large ($>10^6$) number of degrees of freedom, which are adjusted through a regression: an objective (cost) function is chosen – generally associated with an error between what the model estimates and the label value – and then the parameters that minimize the cost function over the entire dataset are chosen to construct the model. Regularization techniques are employed to prevent the model from overfitting the data. This can be done by increasing the cost of complex models or stimulating the discovery of simple models during training.

Here, a convolutional neural network (CNN) is used to model the input–output relationship. The input to the CNN is a reduced image frame written as a matrix where each element corresponds to the luminosity intensity of its corresponding position, *i.e.* a matrix of picture elements (pixels). The output is a single scalar value corresponding to the input's associated heat flux value. CNNs take advantage of the spatial distribution of information in images to assist in modeling data from images. For instance, because image features (e.g., a bubble) present hierarchical structures (edges, contours, bubbles), only the intensity of local neighboring pixels are weighted together in the feedforward process of transforming an input into an output.

A machine learning model is considered "deep" when the number of nested mathematical operations (often named layers) between the input and output is large. This type of neural network architecture has shown success in quantifying heat flux from visualization in a previous publication by our group [16].

The deep convolutional neural network architecture that was used in this study is schematically shown in **Figure S3**. Its general structure consisted of a series of two convolutional layers (stride 1, window size 3) followed by maxpooling and dropout operations, which are repeated three times, and then followed by a fully connected (dense) layer and the scalar output. The maxpool layers are used to progressively reduce by 2^2 the dimensionality of the information as the network progresses forward. The entire model had $\sim 6.6 \times 10^6$ degrees of freedom. The dropout operations are used as a regularization measure, and have rates of 0.2, 0.2, and 0.5 [32]. The model was trained with a mean square error (MSE) cost function in batches of 64 data points, using the Adam minimizer over 100 epochs (meaning each training data point could be seen up to 100 times by the optimizer), with early stopping if the validation cost function increased over consecutive epochs. The machine learning model was implemented in Python using Keras [33] with a Tensorflow backend [33]. The reader is encouraged to visit Ref. [34] for more detailed information on convolutional neural networks and the specific methods used herein.

4. Results

Our results focus on evaluating two key metrics: 1) ability to interpolate visualization data (e.g., predict heat flux from which the CNN has not been trained); and 2) overall precision and accuracy, which we quantify by the distribution of the mean absolute and mean absolute percentage errors (MAE and MAPE, respectively) of a series of predictions for a given experimentally-measured heat flux q'' and their CNN-predicted counterparts q''_{CNN}, *i.e.*,

$$MAE = \left|q'' - q''_{CNN}\right| MAPE = \left|(q'' - q''_{CNN})/q''\right|. \quad (1)$$

To better address each metric, we subdivide our data into two subsets, each of which are further subdivided for training, validation and testing, as is typical in model training (Table 1). In Subset 1 we investigate the ability of a neural network to interpolate and extrapolate boiling heat transfer data. In Subset 2 we incorporate more training data, and demonstrate that it is possible to further increase the precision and accuracy of our CNN model, particularly at high heat flux. Using the latter we explore how CNNs can learn complex data from visualization and the implications for more complex boiling heat transfer scenarios, such as boiling in the unstable transition regime, *i.e.*, during the reestablishment of nucleate boiling after the progression to film boiling. The heat fluxes not selected for each study were not used for training, but were later used for validation and testing.

4.1. Interpolating visualization data for heat flux estimation (Subset 1)

The CNN with architecture shown in **Figure S1** was trained with data corresponding to the heat fluxes shown in Table 1 (Subset 1). We deliberately staggered training heat fluxes in our dataset to investigate whether the trained machine learning model would be capable of interpolating and extrapolating heat fluxes – *i.e.*, to estimate the staggered heat fluxes for which it was not trained. After training, the model's heat flux prediction performance was tested using both training and validation (non-training) data, as measured by MAE and MAPE (**Eq. (1)**. The heat flux estimated by the CNN (q''_{CNN}) *vs.* the heat flux measured by the thermocouples (q'') are shown in Fig. 2**a**. Ideally, the relationship between these quantities should be 1:1 (solid line), representing a perfect correlation. Interestingly, Fig. 2**a** shows that the trained model can predict the trend in increasing heat flux even in data not used for training, with $R^2 > 0.96$. However, at medium-to-high heat fluxes, noticeable scatter in the estimations can be observed. These are better visualized in Fig. 2**b** where the distributions of error deviations are shown as violin plots for low and high heat fluxes, respectively. At very low heat fluxes (<10 W/cm^2), the model is unable to accurately predict

Table 1
Heat flow rates used for training purposes in each study (O indicates used, – indicates unused), with an indication of what is referred to as "low", "medium" and "high" heat flow rates.

Q [W]	Subset 1	Subset 2	
20	–	–	Low
40	O	–	
60	–	O	
80	O	–	
100	–	–	
120	O	O	Medium
150	–	–	
200	O	O	
300	–	O	
400	O	O	
500	–	O	High heat flux
600	O	O	
700	–	O	
800	O	O	
900	–	O	

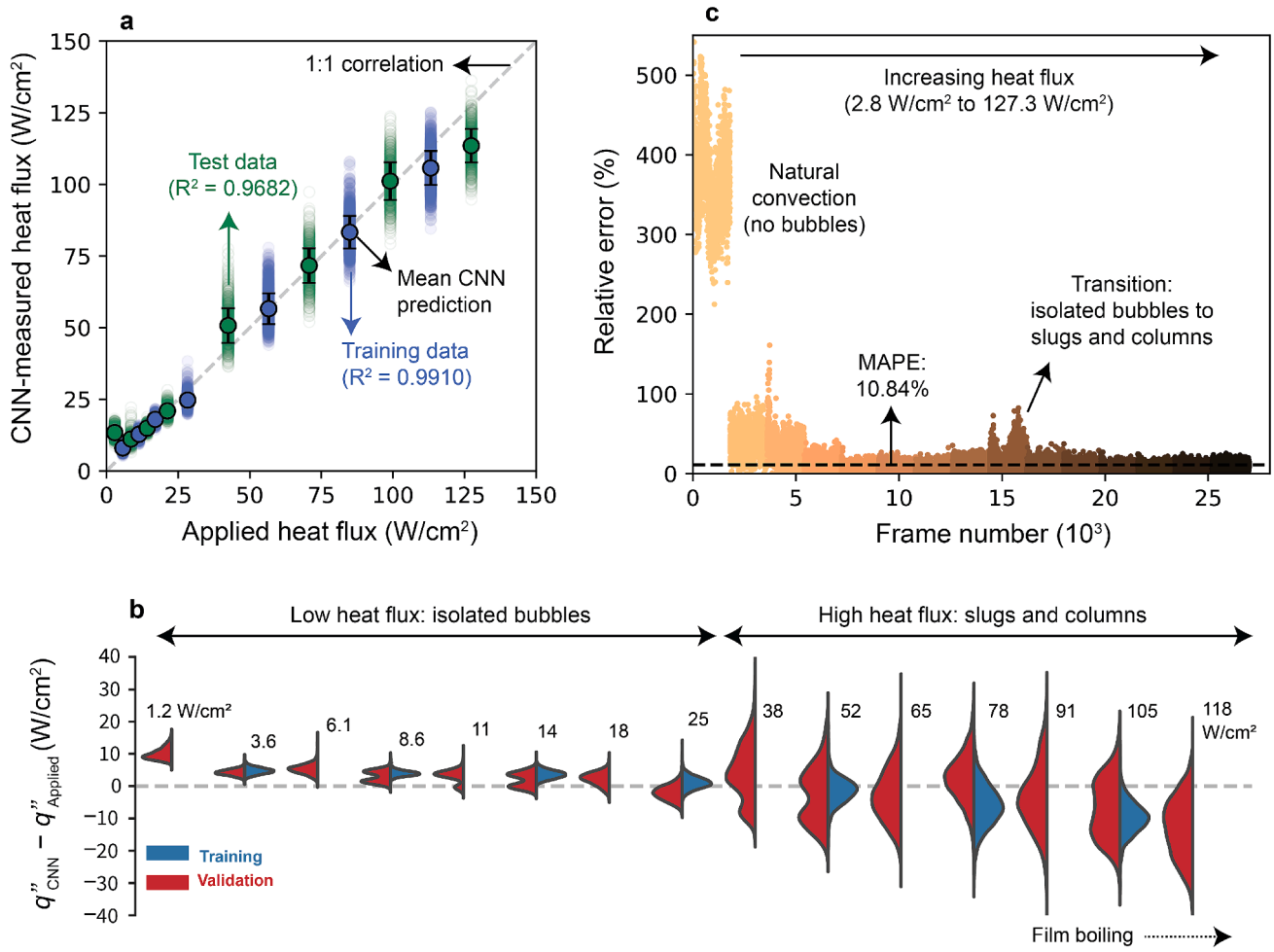

Fig. 2. Metrics of CNN performance trained from Subset 1: interpolating heat fluxes from visualization. (a) Heat flux estimated by the CNN *vs.* heat flux measured by thermocouples using data from 5 distinct experimental tests. (b) Violin plots of the distribution of the deviation between the estimated and measured heat fluxes for low (isolated bubbles) and high (slugs and columns) heat fluxes in training and non-training (e.g., test and validation) data. (c) Error *vs.* image number for all frames in a representative experiment, for varying heat fluxes.

heat flux, and the error distribution does not intercept zero error. At higher heat fluxes (>10 W/cm^2), however, the CNN estimation error intercepts zero, although the error distribution widens considerably. However, despite larger absolute errors at high fluxes, the relative percentage errors under these conditions are much smaller than at low heat flux values. It is worth mentioning that for some heat flux values, the violin-like plot shows a bimodal distribution. This is arguably related to the fact that frames in the same video for an experimental condition present similar biases (e.g. lighting and camera positions) with each mode in the distribution representing one experimental run.

The comparison between the accuracy of the CNN for low and high heat fluxes can be more directly observed in Fig. 2**c**, where the relative percentage errors are plotted for all frames of a representative experiment. Clearly, these percentage errors are orders of magnitude larger at very low heat flux values (2.83 and 5.66 W/cm^2). At 8.49 W/cm^2, however, the error decreases substantially and continues to decrease as heat flux increases, down to ~ 11 %. The lower percentage errors at high heat fluxes is mainly due to the use of a mean squared error cost function in training the CNN (more details can be seen in the Methods section of the Supplementary Material), which penalizes high absolute errors, making the model focus on high heat fluxes, since they present a considerably larger source of cost for model training. Another contributing factor is that at small thermal loads, the degree of overheating is not sufficiently high to allow intense bubble dynamics, as illustrated in Fig. 1**c**. The lack of significant bubble activity in the visualization window makes it intrinsically difficult for a visualization-based algorithm to accurately correlate image data to the thermal load. The discussion above can be better visualized in **Figure S4** and Supplementary Movie 1, which displays the visualization window for different heat fluxes, the MAE and MAPE between the measured and CNN estimate heat fluxes are shown.

Given that the ability of the CNN to predict heat flux with reasonable MAE and MAPE (~<10 W/cm^2 and ~ 11 %, respectively) particularly at high heat flux and for conditions to which it has not been during training, it is possible that it encodes features relevant to boiling heat transfer physics. These will be explored later in this paper.

4.2. Accurate heat flux estimation from visualization (Subset 2)

We now explore whether MAE and MAPE can be improved with a larger dataset, particularly at high heat fluxes. The same CNN architecture (**Figure S3**) was therefore trained almost exclusively on medium and high heat flow rate data (Table 1, Study 2). Note that low heat flow rates were not used in an attempt to keep a uniform distribution of heat flux on the training dataset and prevent the model from being biased by low heat flux values that carry little information in their visualization

window.

Fig. 3**a** shows the heat flux predicted by the CNN, *i.e.*, q''_{CNN} as a function of the image, or frame, number for 7 distinct experiments. For each experiment, the experimentally-measured heat flux (in color scale) increases in unit steps as a function of the image number, and so does the CNN prediction, as expected. The error between the CNN prediction and the measured heat flux is shown in Fig. 3**b**. In general, at low heat fluxes, the CNN model was observed to overestimate its heat flux prediction, whereas at high heat flux the opposite was observed, *i.e.,* the heat flux was underestimated by the CNN, for the same reasons noted in the previous section. At high heat fluxes, however, while the absolute error is larger, the relative percentage error decreases with increasing heat flux.

It is instructive to evaluate how the trained model in this data-rich study compares to the performance of the CNN trained in the previous section. The comparison is shown as errors in heat flux for both Subset 1 and Subset 2 plotted as violin plots as a function of the heat flux in Fig. 3**c**. The precision (width of the error distribution) and accuracy (deviation of the mean with respect to zero) are substantially improved in Subset 2, in particular at higher loads. Future use of larger datasets is expected to further improve upon these metrics. Therefore, our results so far, in both wire-like heaters [16] and now in flat surfaces, suggest that low MAE and MAPE in estimating heat flux can be achieved using data-driven techniques that learn from visualization, can be effectively achieved with existing machine learning models, and in particular CNNs.

4.3. Can we learn how CNNs interpret the boiling phenomenon?

Although interpolation and low MAE/MAPE should be achievable with current models, it is still to date unclear how CNNs correlate visualization data with a heat flux estimate. In particular, it is unclear whether bubble dynamics, which are important to boiling heat transfer physics [24], are correctly captured by these models. Understanding the features identified by the CNN help move towards the tailoring of more general models that enable generalization over multiple operating conditions (e.g., heater geometry, fluid, surface wetting, etc).

To do so, here we computed (1) activation maps and (2) output gradients [34]. The first focuses on analyzing the regions of any given frame that activate the neurons in the trained neural network. These activation maps reveal regions of an input frame that are perceived as relevant by a specific neuron in the CNN to the estimation of the output (*i.e.,* heat flux) [34]. For boiling heat transfer, therefore, we would expect regions of the image where there is significant bubble activity to be more activated, which we have also shown to be the case in previous studies in wire heaters [16]. The activations for all 64 filters in the first convolutional layer of our network (refer to **Figure S3** for the full architecture) are shown in Fig. 4**a** for a randomly selected frame of Subset 2 listed in Table 1. In Fig. 4**a, 17** of the activation filters are uniform (*i.e.*, completely white), implying that these layers are "dead" and do not contribute meaningful information to the heat flux estimation. The majority of the other filters, however, display a strong activation at the interfaces between the bubbles and the background, suggesting that bubble morphology and/or size has a significant effect on heat transfer prediction. Nevertheless, and even though some regions of the image

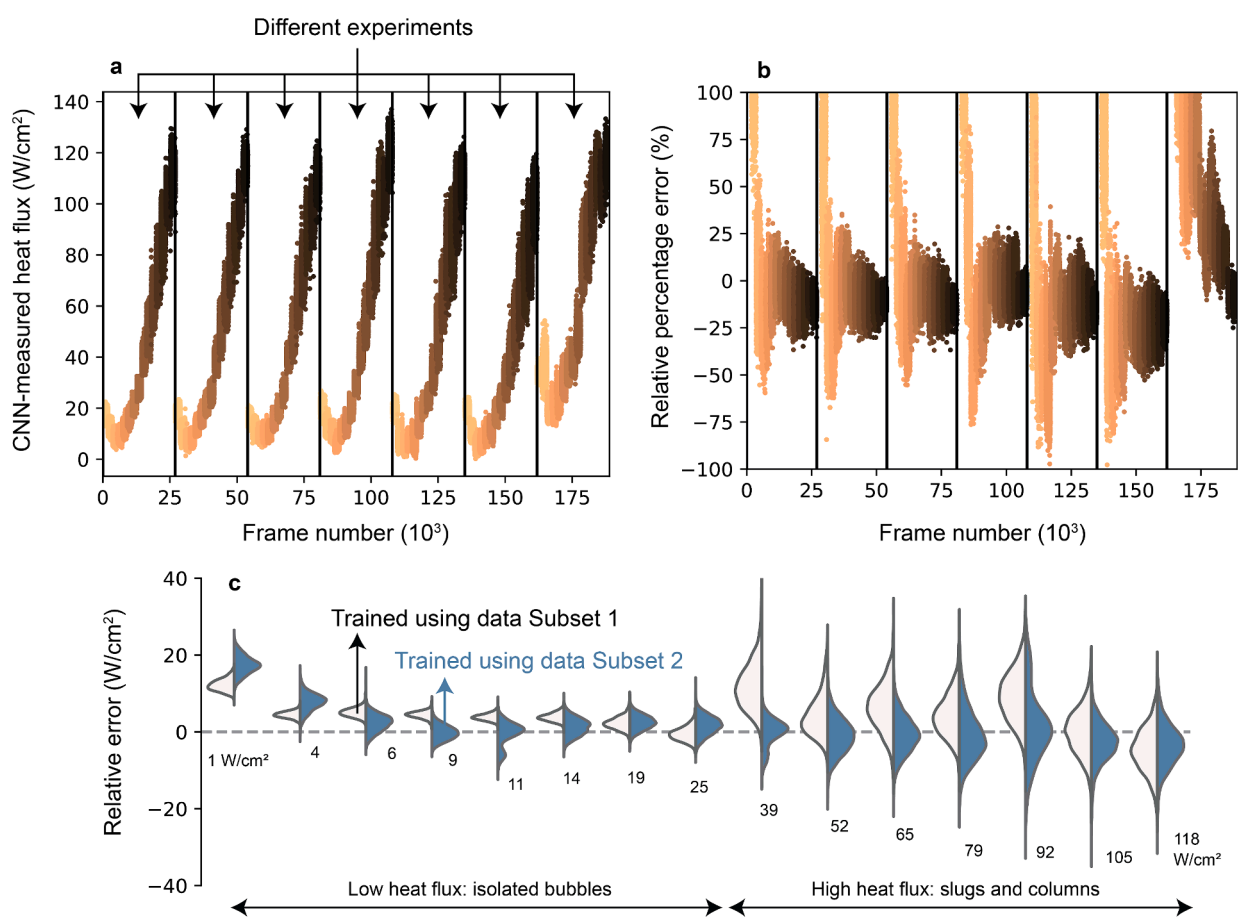

Fig. 3. Metrics of CNN performance trained from Subset 2: interpolating heat fluxes from visualization. (a) Heat flux estimated by the CNN trained in Study 2 for all experimental tests. (b) Accompanying percentage errors for each frame. (c) Comparison between the error distributions in Study 1 and Study 2 for all heat flux values investigated in this paper.

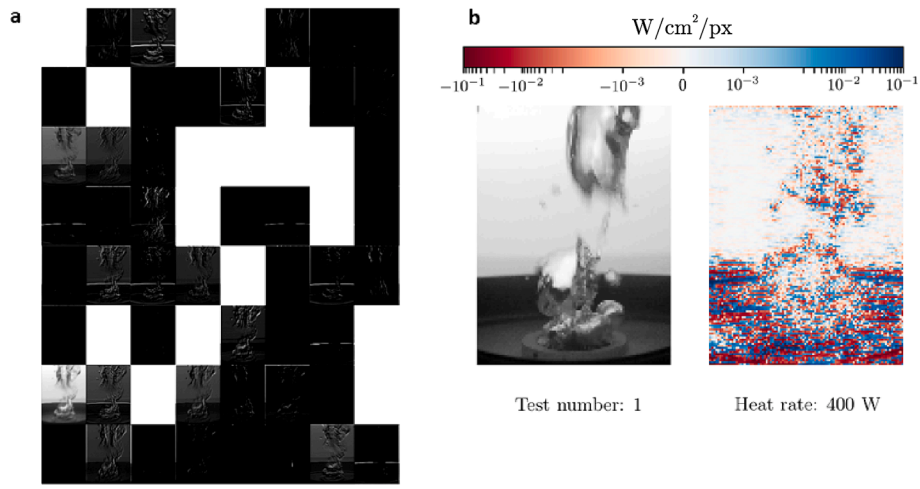

Fig. 4. Frame filtering and pixel sensitivity of the trained CNN. (a) Activations of the first layer of the CNN trained in Subset 2 when looking at a specific frame. Strong gradients in these activations depict regions of the image to which the CNN is sensitive. (b) Selected frame with a heat flow rate of 400 W and corresponding heat flux gradient (W/cm^2/px) plot for each pixel. The gradient plot shows the importance of each pixel on the CNN output. The color scale is linear in the -2×10^3 to 2×10^3 range and logarithmic otherwise.

seem to be contributing the CNN interpretation, it is difficulty pinpoint specific aspects of the image that are actually relevant. Perhaps a more instructive approach would be to calculate the gradient of the predicted heat flux with respect to each pixel's intensity, leading to an image of the same size as the original input (Fig. 4**b**). In this case, regions of strong gradient contribute more heavily to the model's output such as the lower of half of the image, which is mostly occupied by the copper block as well as the liquid–vapor interface. Supplementary Movie 2 shows a similar comparison for the first few seconds of every heat flow rate for every test. Gradient values were found to be stronger near bubbles edges, implying that their form and shape are important for the trained CNN's prediction.

4.4. Towards measurement in unstable regimes

An immediate useful practical application of visualization-based measurement techniques for boiling heat transfer is the ability to infer heat flux much faster than invasive methods that require the system to reach a steady state to be able to infer thermal loads in a Fourier-like way. This is particularly critical for heaters that possess a large thermal capacity. In these, the response time for temperature and heat flux variations to propagate into the measurement instrument (e.g., thermocouple or fluxmeter) is much larger (~seconds) than the timescales intrinsic to the boiling bubble dynamics (~milliseconds), delaying the detection of unsteady boiling events. One of these critical phenomena is the transition to film boiling during regular operation, which is known to be fast and to potentially lead to catastrophic heat transfer degradation, significantly decreasing the thermal load that can be effectively sustained over a safe surface superheat condition. Fig. 5 shows the experimentally-measured heat flux using thermocouples and by applying Fourier's law for three different transient tests, along with the CNN measurement. At time $t = 0$, the total thermal load dissipated by the heaters is 1000 W, which is higher than the one needed to reach critical heat flux. After some time, transient boiling in the transition regime takes place and when a threshold temperature is reached, a safety system is triggered in our experiments and the heater is automatically turned off. According to Fig. 5, the heat flux measured by steady-state Fourier's law drops suddenly. However, this measurement is not reliable because it neglects the substrate's thermal capacitance. Under these conditions, the prediction between our machine learning model and the heat flux measured using Fourier's law were observed to differ significantly. In particular, Fourier's law-based measurement systems are inadequate to be used under transient conditions given the

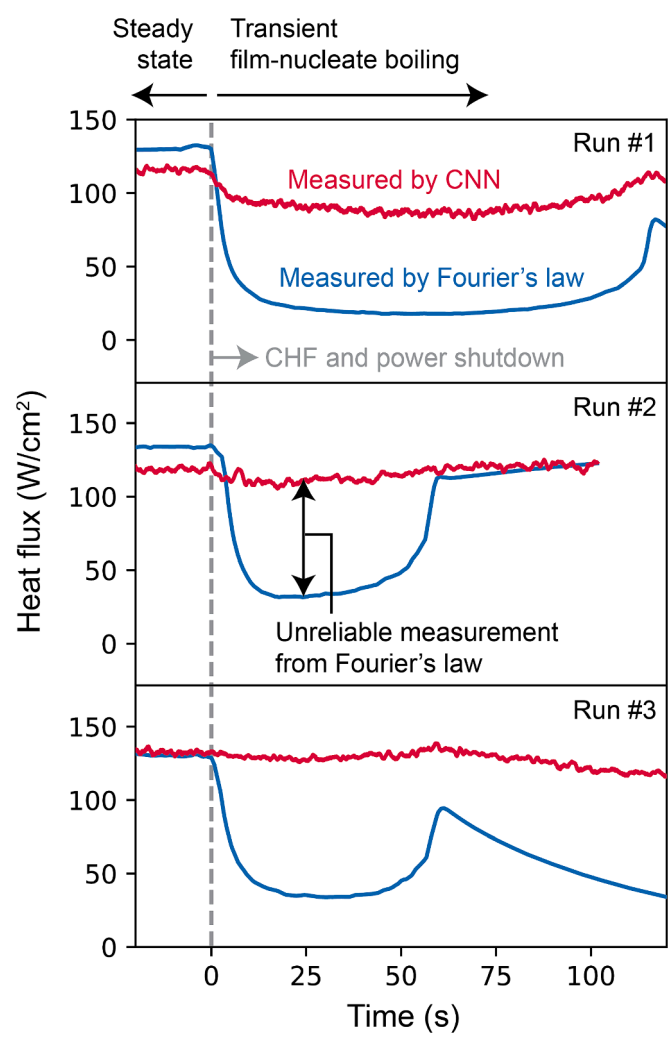

Fig. 5. Heat flux estimation in transient boiling. Heat flux as a function of time as film boiling conditions are reached and the cartridge heaters are turned off using Fourier's law (blue), and using machine learning (red) for three separate tests.

implicit assumption that the system be at steady-state (so that $dT/dt\ k\nabla^2 T\ 0$). In fact, even assuming a modest heat flux coefficient during transition boiling of ~ 10 kW/(m^2 K), the thermal time constant of the heater block is $M\ c_p/(hA) = \sim 50$ s, and thus the heat flux to the liquid pool is expected to be constant for $\gg 10$ s, unlike indicated by Fourier's law. Interestingly, however, the CNN-measured values remain consistently constant even after power is shut down because its output relies exclusively on the bubble dynamics and not on the thermal inertia of the substrate. Consequently, a CNN trained with actual transition boiling images might be a more accurate to estimate a transient heat flux at the surface. Nonetheless, further studies are needed to validate the accuracy of unsteady heat flux predictions, though direct validation is currently challenging due to the experimental difficulty in measuring fast transient heat fluxes such as those seen in transition boiling. In this context, machine learning/computer vision diagnostics of boiling heat transfer may provide a viable pathway to transient heat flux measurements at high heat flux, which is largely unattainable by conventional instruments that rely on thermal equilibrium.

5. Conclusion

Here we investigated whether a combined computer vision-machine learning approach using a convolutional neural network (CNN) is capable of quantifying pool boiling heat flux on a horizontal, flat circular heater based exclusively on visualization frames of the boiling process and without reliance on thermohydraulic parameters. Our experimental setup was designed such that the heat flux can be directly measured from the temperature of the heater block. A total of 15 heat flux levels were imposed to the heater, which altogether encompass all three boiling regimes: natural convection, nucleate boiling and transition boiling. For repeatability purposes and, to avoid any bias, a total of 7 different sets of tests were performed with image data being recorded by a conventional DSLR camera along with the heater's temperature measurements.

Our results showed that boiling heat flux can be measured from visualization with low error ($\sim< 10$ W/cm^2, $\sim< 11$ %), particularly at higher heat fluxes (*i.e.*, boiling regimes with higher bubble frequency departure). Our analysis also revealed that the CNN model can interpolate images and measure heat flux values for which it had not been trained. Different, and given its exclusive reliance on visualization data, we also shown that the visualization-based quantification might present lower accuracy for transient boiling regimes if the algorithm is not trained with images of that specific regime. The performance for the current CNN is expected to be dependent on many of the biases in the dataset including the viewing angle, lighting conditions, geometric characteristics, and working fluid. Although the CNN cannot be readily used for other conditions, the presented methodology can potentially be applicable. Given the limitations previously discussed, efforts towards the generalization of these image-based, machine-learning assisted diagnostic tools certainly is an area worth investigation for future studies (e.g., different working fluids). Finally, it would be interesting to investigate boiling surface that are significantly larger than the Taylor's wavelength, making the phase change process independent of the surface's dimension.

CRediT authorship contribution statement

Vinicius K. Scariot: Conceptualization, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – review & editing. **Gustavo M. Hobold:** Conceptualization, Formal analysis, Writing – original draft. **Alexandre K. da Silva:** Formal analysis, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.applthermaleng.2024.123068.

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